

$$\begin{aligned}
I(-4, 0, -2, -2) &= q^5 \eta^{v_0} \eta^{3v_1} \\
I(-4, 0, -2, -1) &= q^5 \eta^{v_0} \eta^{3v_1} \\
I(-4, 0, -2, 0) &= q^5 \eta^{v_0} \eta^{3v_1} \\
I(-4, 0, -2, 1) &= q^5 \eta^{2v_0} \eta^{4v_1} \\
I(-4, 0, -2, 2) &= q^5 \eta^{3v_0} \eta^{5v_1} \\
I(-4, -2, -1, -2) &= -q^5 \eta^{v_0} \eta^{3v_1} + q^5 \eta^{2v_0} \eta^{4v_1} \\
I(-4, -2, -1, -1) &= -q^5 \eta^{v_0} \eta^{3v_1} + q^5 \eta^{2v_0} \eta^{4v_1} \\
I(-4, -2, -1, 0) &= q^5 \eta^{2v_0} \eta^{2v_1} - q^5 \eta^{3v_0} \eta^{3v_1} \\
I(-4, 0, -1, -2) &= q^4 \eta^{4v_1} + q^4 \eta^{v_0} \eta^{3v_1} - q^5 \eta^{-v_0} \eta^{3v_1} + 2q^5 \eta^{4v_1} + q^5 \eta^{v_0} \eta^{3v_1} \\
I(-4, 0, -1, -1) &= q^4 \eta^{4v_1} + q^4 \eta^{v_0} \eta^{3v_1} - q^5 \eta^{-v_0} \eta^{3v_1} + 2q^5 \eta^{4v_1} + q^5 \eta^{v_0} \eta^{3v_1} \\
I(-4, 0, -1, 0) &= q^4 \eta^{4v_1} + q^4 \eta^{v_0} \eta^{3v_1} + q^5 \eta^{v_0} \eta^{5v_1} + q^5 \eta^{2v_0} \eta^{4v_1} \\
I(-4, 0, -1, 1) &= q^4 \eta^{v_0} \eta^{5v_1} + q^4 \eta^{2v_0} \eta^{4v_1} - q^5 \eta^{v_0} \eta^{3v_1} + q^5 \eta^{v_0} \eta^{5v_1} + 2q^5 \eta^{2v_0} \eta^{4v_1} \\
I(-4, 0, -1, 2) &= q^4 \eta^{2v_0} \eta^{6v_1} + q^4 \eta^{3v_0} \eta^{5v_1} - q^5 \eta^{2v_0} \eta^{4v_1} + q^5 \eta^{2v_0} \eta^{6v_1} + 2q^5 \eta^{3v_0} \eta^{5v_1} \\
I(-4, +2, -1, 0) &= q^5 \eta^{-3v_0} \eta^{3v_1} - q^5 \eta^{-2v_0} \eta^{4v_1} \\
I(-4, +2, -1, 1) &= -q^5 \eta^{-v_0} \eta^{3v_1} + q^5 \eta^{4v_1} \\
I(-4, +2, -1, 2) &= -q^5 \eta^{4v_1} + q^5 \eta^{v_0} \eta^{5v_1} \\
I(-4, -4, 0, 0) &= q^4 \eta^{2v_0} \eta^{2v_1} - q^4 \eta^{3v_0} \eta^{3v_1} + q^5 \eta^{v_0} \eta^{v_1} - q^5 \eta^{2v_0} \eta^{2v_1} - q^5 \eta^{3v_0} \eta^{3v_1} + q^5 \eta^{4v_0} \eta^{4v_1} \\
I(-4, -2, 0, -2) &= q^5 \eta^{v_0} \eta^{3v_1} - q^5 \eta^{2v_0} \eta^{4v_1} \\
I(-4, -2, 0, -1) &= q^4 \eta^{v_0} \eta^{3v_1} - q^4 \eta^{2v_0} \eta^{4v_1} + q^5 \eta^{v_0} \eta^{3v_1} - 2q^5 \eta^{2v_0} \eta^{4v_1} + q^5 \eta^{3v_0} \eta^{5v_1} \\
I(-4, -2, 0, 0) &= q^3 \eta^{v_0} \eta^{3v_1} - q^3 \eta^{2v_0} \eta^{4v_1} + q^4 \eta^{2v_1} - q^4 \eta^{v_0} \eta^{3v_1} - q^4 \eta^{2v_0} \eta^{4v_1} + q^4 \eta^{3v_0} \eta^{5v_1} + q^5 \eta^{-v_0} \eta^{v_1} \\
I(-4, -2, 0, 1) &= q^4 \eta^{2v_0} \eta^{4v_1} - q^4 \eta^{3v_0} \eta^{5v_1} + q^5 \eta^{2v_0} \eta^{4v_1} - 2q^5 \eta^{3v_0} \eta^{5v_1} + q^5 \eta^{4v_0} \eta^{6v_1} \\
I(-4, -2, 0, 2) &= q^5 \eta^{3v_0} \eta^{5v_1} - q^5 \eta^{4v_0} \eta^{6v_1} \\
I(-4, 0, 0, -2) &= q^2 \eta^{4v_1} + q^5 \eta^{-v_0} \eta^{3v_1} - 3q^5 \eta^{4v_1} + q^5 \eta^{v_0} \eta^{5v_1} \\
I(-4, 0, 0, -1) &= q^2 \eta^{4v_1} + q^4 \eta^{-v_0} \eta^{3v_1} - 2q^4 \eta^{4v_1} + q^4 \eta^{v_0} \eta^{5v_1} + 2q^5 \eta^{-v_0} \eta^{3v_1} - 4q^5 \eta^{4v_1} + 2q^5 \eta^{v_0} \eta^{5v_1} \\
I(-4, 0, 0, 0) &= q^2 \eta^{4v_1} + q^3 \eta^{-v_0} \eta^{3v_1} - 2q^3 \eta^{4v_1} + q^3 \eta^{v_0} \eta^{5v_1} + q^4 \eta^{-2v_0} \eta^{2v_1} - 3q^4 \eta^{4v_1} + 2q^4 \eta^{v_0} \eta^{5v_1} + \\
I(-4, 0, 0, 1) &= q^2 \eta^{v_0} \eta^{5v_1} + q^4 \eta^{4v_1} - 2q^4 \eta^{v_0} \eta^{5v_1} + q^4 \eta^{2v_0} \eta^{6v_1} + 2q^5 \eta^{4v_1} - 4q^5 \eta^{v_0} \eta^{5v_1} + 2q^5 \eta^{2v_0} \eta^{6v_1} \\
I(-4, 0, 0, 2) &= q^2 \eta^{2v_0} \eta^{6v_1} + q^5 \eta^{v_0} \eta^{5v_1} - 3q^5 \eta^{2v_0} \eta^{6v_1} + q^5 \eta^{3v_0} \eta^{7v_1} \\
I(-4, +2, 0, -2) &= q^5 \eta^{-3v_0} \eta^{3v_1} - q^5 \eta^{-2v_0} \eta^{4v_1} \\
I(-4, +2, 0, -1) &= q^4 \eta^{-3v_0} \eta^{3v_1} - q^4 \eta^{-2v_0} \eta^{4v_1} + q^5 \eta^{-3v_0} \eta^{3v_1} - 2q^5 \eta^{-2v_0} \eta^{4v_1} + q^5 \eta^{-v_0} \eta^{5v_1} \\
I(-4, +2, 0, 0) &= q^3 \eta^{-3v_0} \eta^{3v_1} - q^3 \eta^{-2v_0} \eta^{4v_1} + q^4 \eta^{-4v_0} \eta^{2v_1} - q^4 \eta^{-3v_0} \eta^{3v_1} - q^4 \eta^{-2v_0} \eta^{4v_1} + q^4 \eta^{-v_0} \eta^{5v_1} \\
I(-4, +2, 0, 1) &= q^4 \eta^{-2v_0} \eta^{4v_1} - q^4 \eta^{-v_0} \eta^{5v_1} + q^5 \eta^{-2v_0} \eta^{4v_1} - 2q^5 \eta^{-v_0} \eta^{5v_1} + q^5 \eta^{6v_1} \\
I(-4, +2, 0, 2) &= q^5 \eta^{-v_0} \eta^{5v_1} - q^5 \eta^{6v_1} \\
I(-4, +4, 0, 0) &= q^4 \eta^{-6v_0} \eta^{2v_1} - q^4 \eta^{-5v_0} \eta^{3v_1} + q^5 \eta^{-7v_0} \eta^{v_1} - q^5 \eta^{-6v_0} \eta^{2v_1} - q^5 \eta^{-5v_0} \eta^{3v_1} + q^5 \eta^{-4v_0} \eta^{4v_1} \\
I(-4, -2, 1, 0) &= q^5 \eta^{v_0} \eta^{5v_1} - q^5 \eta^{2v_0} \eta^{6v_1} \\
I(-4, -2, 1, 1) &= q^5 \eta^{2v_0} \eta^{6v_1} - q^5 \eta^{3v_0} \eta^{7v_1} \\
I(-4, -2, 1, 2) &= q^5 \eta^{3v_0} \eta^{7v_1} - q^5 \eta^{4v_0} \eta^{8v_1} \\
I(-4, 0, 1, -2) &= -q^5 \eta^{-v_0} \eta^{5v_1} - q^5 \eta^{v_0} \eta^{5v_1} \\
I(-4, 0, 1, -1) &= -q^5 \eta^{-v_0} \eta^{5v_1} - q^5 \eta^{v_0} \eta^{5v_1} \\
I(-4, 0, 1, 0) &= -q^5 \eta^{6v_1} - q^5 \eta^{v_0} \eta^{5v_1} \\
I(-4, 0, 1, 1) &= -q^5 \eta^{v_0} \eta^{5v_1} - q^5 \eta^{v_0} \eta^{7v_1} \\
I(-4, 0, 1, 2) &= -q^5 \eta^{2v_0} \eta^{6v_1} - q^5 \eta^{2v_0} \eta^{8v_1} \\
I(-4, +2, 1, -2) &= q^5 \eta^{-2v_0} \eta^{4v_1} - q^5 \eta^{-v_0} \eta^{5v_1} \\
I(-4, +2, 1, -1) &= q^5 \eta^{-2v_0} \eta^{4v_1} - q^5 \eta^{-v_0} \eta^{5v_1} \\
I(-4, +2, 1, 0) &= q^5 \eta^{-2v_0} \eta^{4v_1} - q^5 \eta^{-v_0} \eta^{5v_1} \\
I(-4, +2, 1, 1) &= q^5 \eta^{-2v_0} \eta^{4v_1} - q^5 \eta^{-v_0} \eta^{5v_1} \\
I(-4, +2, 1, 2) &= q^5 \eta^{-2v_0} \eta^{4v_1} - q^5 \eta^{-v_0} \eta^{5v_1}
\end{aligned}$$